GY462—Real Estate Finance

Lecture 6

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Topics of This Lecture

- Introduction to mortgage basics & analysis
  - Fixed rate mortgage loans and basic formulas
- Look at applications and the concept of 'effective cost of borrowing'

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Contents

Fixed Rate Mortgage Loans

- Basic Formula
- Applications: IO, CAM, CPM
- Effective Cost of Borrowing
  - Discount Points
  - Prepayment Penalties
  - (Yield Maintenance)

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Mortgage Basics: Compounding Process

- Key idea: Earn <u>interest on interest</u>
- If (usually) compounded <u>monthly</u>
  - Nominal yield of 6% per annum
  - Year 1: $100\ (1+6\%/12)^{12} = 106.17$
  - Effective Annual Yield (EAY) = 6.17%
- Convention
  - *Nominal yield = Effective monthly yield × 12*

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Fixed Rate Mortgage Loans

- **Basic formula**
  - **Balance:** *BAL*
  - **Payment:** *PMT*
  - **Amortization:** $AMORT_t = BAL_{t-1} - BAL_t$
  - $BAL_t = (1+i)\ BAL_{t-1} - PMT_t$

  ⇒ **That is BAL_t is interest on previous balance + the balance itself minus the payment…**
  ⇒ **Interest rate is 'cost of principal'**
  ⇒ **What is the payment?**

Fixed Rate Mortgage Loans – *Continued*

- **Defining the payment**
  - **Using above formulas…**

  ⇒ $PMT_t = (1+i)\ BAL_{t-1} - BAL_t$ **(1)**

  ⇒ $PMT_t = AMORT_t + i \times BAL_{t-1}$ **(2)**

- **The payment always consists of…**
  - **Cost of principal (interest on balance)**
  - **Amortization of loan**

Fixed Rate Mortgage Loans – *Remarks*

- **Some key concepts**
  - **Amortization always refers to capital reimbursement**
  - **Payment is composed of amortization + interest on the outstanding balance**
  - **Total loan, i.e. balance at time *t* (BAL_t) is discounted sum of future payments**
  - **Like an annuity paying PMT_t (with payment that might change in amount)…**
  - **Can you see why?**

Computing Remaining Balance

Reformulate (1):

$$BAL_t = \frac{PMT_{t+1}}{(1+i)} + \frac{BAL_{t+1}}{(1+i)} \quad \text{and}$$

$$BAL_{t+1} = \frac{PMT_{t+2}}{(1+i)} + \frac{BAL_{t+2}}{(1+i)} \quad \text{and} \ \ldots\ \ldots\ \ldots\ BAL_N = 0$$

$$\Rightarrow \quad BAL_t = \frac{PMT_{t+1}}{(1+i)} + \frac{PMT_{t+2}}{(1+i)^2} + \frac{BAL_{t+2}}{(1+i)^2}$$

$$\Rightarrow \quad BAL_t = \frac{PMT_{t+1}}{(1+i)} + \frac{PMT_{t+2}}{(1+i)^2} + \ldots + \frac{PMT_N}{(1+i)^{N-t}}$$

$$\Rightarrow \quad BAL_t = \sum_{k=1}^{N-t} \frac{PMT_{t+k}}{(1+i)^k} = \text{PV of remaining payments}$$

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Computing Remaining Balance — *Example*

- **30-year fixed rate mortgage**
  - **Compounded <u>monthly</u>** *(N = 360)*
  - **Annual interest rate: 9.125%**
  - **Loan amount: £120,000**
  - **<u>Constant payment & fully amortizing</u>**
  - **First payment: April 2026**

Computing Remaining Balance — *Example*

- **What is the monthly payment?**
- **What is the remaining balance at the end of year 2028?**
- **How much principal will you have paid back until the end of 2028?**
- **How much interest will you need to pay until the end of 2028?**

Computing Remaining Balance — *Manually…*

- **Could do it manually…**
- **PMT is the annuity such that…**

$$\sum_{t=1}^{360} \frac{PMT_t}{\left(1+\frac{9.125}{12}\right)^t} = £120000$$

- **Solve this and find that payment is 976.36 monthly…**
- **What are the next steps?**

Computing Remaining Balance — *Manually…*

- **BAL(1) = 120,000 + 120,000 (0.76) – 976.36 = 119,936.14**
- **AMORT(1) = BAL(1) - BAL(0) = …**
  - **119,936.14 - 120,000 = 63.86**
- **Balance, payment and amortization at end of 2028?**
  - **Monthly compounding, so repeat calculations for 33 steps**
  - **9 Months of 2026 + 24 months till end of 2028**

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Computing Payment with Your Calculator… (Illustration based on TI BA II Plus)

Key: Use Time Value of Money Worksheet

2nd [RESET] ENTER	(Reset memory)
2nd [P/Y] 12 ENTER	(Set year PMT)
2nd [QUIT]	(Change function)
30 2nd [xP/Y] N	(Enter n. of payments)
9.125 I/Y	(Nominal interest rate)
120000 +/- PV	(Borrowed amount)
CPT PMT	(Compute payments)

Display: 976.36*

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Computing Remaining Balance with Your Calculator…

Key: Use Amortization Worksheet

2nd [Amort]

(Select the Amortization worksheet)

↓ 33 ENTER

↓

Display BAL: -117,614.86*

↓

Display PRN: 2385.14*

↓

Display INT: 29,834.74*

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'Focus point 1'

Mortgage calculations: payments, amortization and interests – all clear?

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Contents

Fixed Rate Mortgage Loans

- Basic Formula
- **Applications: IO, CAM, CPM**
- Effective Cost of Borrowing
  - Discount Points
  - Prepayment Penalties
  - Yield Maintenance

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Types of Fixed Rate Mortgage Loans

1. **Interest Only Loan (IO)**
2. **Constant Amortization Mortgage (CAM)**
3. **Constant Payment Mortgage (CPM)**

Interest Only Loan

- **By definition** $AMORT_t = 0$
  - $PMT_t = i \times BAL_{t-1}$
  - $BAL_t = BAL_{t-1}$
  - **At the end:** $PMT_N = (1+i)\ BAL_{N-1}$

⇨ **Payment pattern: Spike at the end**

⇨**Do borrowers like IO loans?**

⇨**Do lenders like IO loans?**

'Focus point 2'

What do you think borrowers like?
And lenders?

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Constant Amortization Mortgage

- **By definition** $AMORT_t = BAL_0 / N$
  - $BAL_t = BAL_0 (1 - t / N)$
  - $PMT_t = i \times BAL_{t-1} + BAL_0 / N$
  - $PMT_t = BAL_0 (i - i(t-1)/N + 1/N)$

⇨ **Declining linear payment**
  - **Contrary to typical income growth expectations**

'Focus point 3'

Using the same data as in the loan above, can you work out the payments?
[Data: 120,000; 30 years/360 months, 9.125% annual]

Constant Payment Mortgage

- **By definition** $PMT_t = PMT$
  - $BAL_t = (1+i)\ BAL_{t-1} - PMT$
  - $AMORT_t = BAL_{t-1} - BAL_t$
  - $AMORT_t = PMT - i \times BAL_{t-1}$

⇨ **Over time, BAL declines, AMORT increases**

- **How does this compare with CAM?**

Comparison of CPM & CAM — *Remarks*

- **CPM: balance declines over time so…**
  - **Interest payment decreases**
  - **To keep PMT constant, AMORT must go up**
- **CAM: First payment above CPM; why?**
  - **Have to reimburse same amount but…**
  - **CPM growing AMORT; here constant!**
  - **First reimbursement must lie above PMT of CPM (so PV of payments is same = LOAN)**

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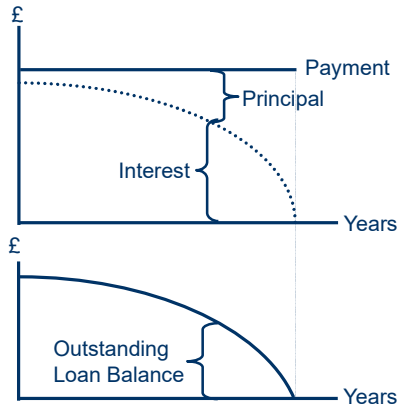
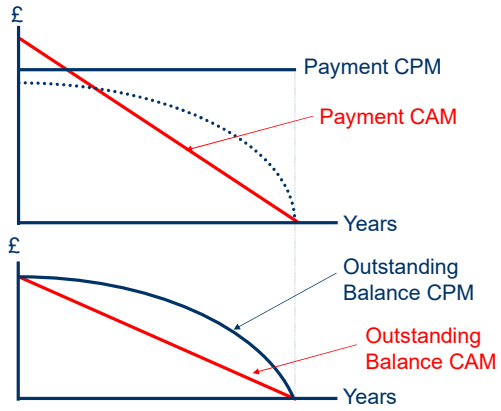
Constant Payment Mortgage — *Graphical Illustration*



Comparison of CPM & CAM —Graphical Illustration



Negative amortization?

- **Lender and borrower can agree on…**
  - **Loan balance at maturity to be bigger than initial loan**
  - **Payments are negotiated at lower level than under simple CPM**
- **In both cases negative amortization**
  - **Solve either to find PMT (first case) or**
  - **To find out final loan balance (second case)**
- ***Take a look at examples in B&F!!!***

Contents

Fixed Rate Mortgage Loans

- Basic Formula
- Applications: IO, CAM, CPM
- **Effective Cost of Borrowing**
  - **Discount Points**
  - **Prepayment Penalties**
  - **(Yield Maintenance)**

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Effective Cost of Borrowing

- **Terminology:**
  IRR on loan = 'Yield to Maturity' (YTM)

⇨ **Question: Is YTM always equal to contractual interest rate?**

Effective Cost of Borrowing — *Continued*

- **Additional costs**
  - **Origination fees in form of 'discount points' (or 'prepaid interest')**
  - **Other origination charges**
  - **Prepayment penalties**

⇨ **Drive wedge between YTM and contractual rate**

Discount Points

- **Fee in the form of discount applied to the loan**
  - **3 discount points on a 100 loan: Borrower only gets 97 but owes 100**

⇨ **Effect: Increases IRR on loan**

Discount Points and Effective Cost — *Example*

- **Consider CPM of 1,000,000**
  - **8% annual interest**
  - **Repaid monthly in 30 years**
  - **1 discount point**
- **YTM (IRR) if no discount points?**
  - **Identical to contractual rate of** *8%/12 = 0.667%*
- **Effective cost of borrowing?**

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Discount Points and Effective Cost — *Example*

- **Effective cost of borrowing?**
  - **Compute PMT no discount points:** *7,337.65*

$$-1{,}000{,}000 = 7{,}337.65/(1+YTM/12)+\ldots+7{,}337.65/(1+YTM/12)^{360}$$

*0.99

  - **Compute** *IRR (YTM)*: *0.667% per month or 8% p.a.*

$$990{,}000 = 7{,}337.65 \times (1/(1+IRR)+\ldots+1/(1+IRR)^{360})$$

  - **Compute** *IRR (YTM)*: *0.6755% per month or 8.11% p.a.*

With the Calculator…

Key: Use Time Value of Money Worksheet

2nd [RESET] ENTER (Sets all variables to defaults)

2nd [P/Y] 12 ENTER (Defines monthly compounding)

CE/C (Exit defining compounding period)

30 2nd [xP/Y] N (Enter number of payments)

8 I/Y (Enter annual interest rate)

1000000 +/- PV (Enter loan amount)

CPT PMT (Computes payment)

Display: *PMT* = 7,337.65 *

With the Calculator…

Key: Select Cash Flow Worksheet Data

2nd [FORMAT] 4 ENTER CE/C (Show 4 decimals)

CF (Select Cash Flow Worksheet)

990,000 +/- ENTER (Enter initial CF)

↓ 7337.65 ENTER

↓ 360 ENTER (Enter CFs for months 1-360)

IRR CPT (Compute IRR (~Yield to Maturity))

Display: *IRR* = 0.6755% * ⇒ *YTM* (0.6755%) > *i* (0.6667%)

'Focus point 4'

Effective cost calculations: all clear?
Is this the only way to do them?

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Early Repayment

- **Now suppose you can repay loan after 5 years**
  - **Compute remaining balance: 950,698.25**

$BAL=7{,}337.65\times(1/(1+0.08/12)+\ldots+1/(1+0.08/12)^{300})$

  - **Compute IRR: 8.25% per annum**

$990{,}000=7{,}337.65\times(1/(1+IRR)+\ldots+1/(1+IRR)^{60})$
$+\ 950{,}698.25/(1+IRR)^{60}$

With the Calculator…

[2nd] [RESET] [ENTER] (Sets all variables to defaults)

[2nd] [P/Y] 12 [ENTER] [CE/C] (Defines monthly compounding)

30 [2nd] [xP/Y] [N] (Enter number of payments)

8 [I/Y] (Enter annual interest rate)

1000000 [+/-] [PV] (Enter loan amount)

[CPT] [PMT] (Computes payment) Display: *PMT* = 7337.65 *

[2nd] [AMORT]

[↓] 60 [ENTER] (Set ending period to 60)

[↓] Display: *BAL* = –950,698.25 *

With the Calculator…

[2nd] [RESET] [ENTER] (Sets all variables to defaults)

[2nd] [FORMAT] 4 [ENTER] (Show 4 decimals)

[CF] (Select Cash Flow Worksheet)

990,000 [+/-] [ENTER] (Enter initial CF)

[↓] 7337.65 [ENTER]

[↓] 59 [ENTER] (Enter CFs for months 1-59)

[↓] 950698.25 [+] 7337.65 [=] [ENTER] (Enter CF for month 60)

[IRR] [CPT] (Compute IRR (~Yield to Maturity))

Display: *IRR* = 0.6875% * ⇒ *IRR p.a.* = 8.25%

'Focus point 5'

Effective cost with prepayment:
are the calculations clear?
Is this the only way to do them?

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Even Earlier Repayment…

- **Suppose repay above loan after 1 year**
  - **Compute remaining balance: 991,646.31**

$BAL=7,337.65\times(1/(1+0.08/12)+\ldots+1/(1+0.08/12)^{348})$

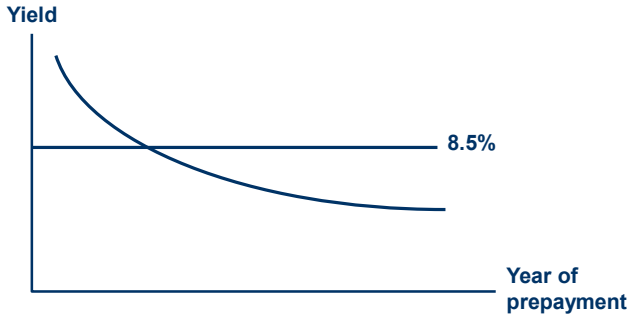
  - **Compute IRR: 9.05% per annum**

$990,000=7,337.65\times(1/(1+IRR)+\ldots+1/(1+IRR)^{12})$
$+ 991,646.31/(1+IRR)^{12}$

⇨ **Difference between *IRR/YTM* & *interest* rate *higher* the *earlier* the loan is repaid**

What Does the Mobile Borrower Prefer?

- **Suppose alternative is *8.5%, no points***



Why do Lenders Impose 'Discount Points'?

- **Way for lender to insure against prepayment risk**
  - **When repaid: Lender needs to invest at uncertain rate (but gets compensated)**
- **Discount points reduce chance to attract 'mobile borrowers'**
  - **Help to get good 'selection of borrowers' (avoid adverse selection)**

Why Are Borrowers Willing to Pay 'Discount Points'?

- **Prepayment has value for borrower**
  - **Option value: Pay back when interest rates fall and borrow at lower rate**
  - **Borrower is willing to pay a premium**
- **Points helps borrower to qualify for loan (lower initial PMT if nominal rate is reduced)**

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'Focus point 6'

Discount points:
what benefits for the lenders?
And for the borrowers?

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Prepayment Protection

- **Prepayment penalty**
  - **Charge to borrower if repay early**
  - **Usually some % of outstanding balance**
  - **Suppose 3% of balance on 5 year example: Raises IRR to 8.72% (with 1 discount point)**

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Outlook — *Lecture 7*

Mortgage Payment Issues

- **The role of inflation**
- **Graduated payments mortgages**
- **Adjustable-rate mortgages**

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